

PROPERTIES OF A NONPARAMETRIC SEQUENTIAL TEST OF MILLER WITH THE VIEW TOWARDS APPLICATIONS

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SUMMARY

Miller (1972) proposed a nonparametric sequential test based on the Wilcoxon one-sample rank statistic to test the hypothesis that $\theta=0$, where θ is the centre of symmetry of a symmetric distribution. In addition to Miller's work some further properties of this test are studied in selected cases by means of Monte Carlo methods. It is shown that this test saves observations in comparison with the corresponding fixed sample Wilcoxon test. To make it easier to compare and evaluate the test for practical use, approximations are given for the operating characteristic function and the average sample number. These approximations are general and can be calculated for different significant levels, truncation points and underlying distributions.

1. The need for a simple nonparametric sequential test

Consider the problem of testing $H_0: \theta = 0$ against a shift alternative, where θ is the centre of symmetry of an unknown distribution. A typical situation where this problem arises is in medical trials when paired observations are used to test the difference between two treatments. Usually there is a scarcity in patients who are readily available for such tests, and for ethical, economical and safety reasons the sample size of such a trial should be kept as small as possible. Especially in cases where the sample suggests far deviations from H_0 , it would be desirable to draw a decision with only a small number of observations on hand. A sequential test seems to be the answer to these requirements, and in particular for this problem: a nonparametric procedure.

For the purpose of testing H_0 , Miller (1972) suggested a very simple test which is based on the Wilcoxon one-sample rank statistic. Very few other sequential tests in this category exist as can be seen in the references made by Miller.

The purpose of this paper is to study some properties of this test in addition to those in Miller's article. Since the question arises why a sequential test is better to use in this particular problem, some comparisons will be made with the fixed sample size Wilcoxon one-sample rank test. To make the Miller test more practically appealing, approxi-

mations to the operating characteristic (OC) function and average sample number (ASN) will be given which can be applied generally for known underlying distributions. The approximation for the OC function is believed to be more accurate than that of Miller's, while ASN-approximations are not given by Miller.

2. Miller's test procedure

Let $X_1, X_2, \dots$ be independent observations from a symmetric population with centre of symmetry θ .

Denote

$$(1) \quad SR_n = \sum_{i=1}^n \sum_{j=1}^i \operatorname{sgn}(X_i + X_j) \\ = SR_{n-1} + \sum_{i=1}^n \operatorname{sgn}(X_i + X_n),$$

where

$$(2) \quad \operatorname{sgn}(X) = \begin{cases} 1 & \text{when } X > 0 \\ 0 & \text{when } X = 0 \\ -1 & \text{when } X < 0 \end{cases}$$

The procedure to test H_0 against the alternative $\theta \neq 0$ at level 2α is:

Choose a suitable truncation point ℓ and take observations $X_1, X_2, \dots$ as long as

$$(3) \quad |SR_n| < W_{\alpha, \ell} n, \text{ or}$$

$$(4) \quad |SR_n| > \ell W_{\alpha, \ell} - \frac{1}{2}(\ell - n)(\ell + n + 1)$$

As soon as one of (3) or (4) does not hold, stop the procedure with the decision to reject or accept H_0 respectively. The value of $W_{\alpha, \ell}$ is determined by the prescribed truncation point ℓ as well as α . For a one-sided alternative, $|SR_n|$ is replaced by SR_n in the criteria (3) and (4), and the level of the test is taken to be α .

Note that the decision-rule of this test is very simple, and that $W_{\alpha, \ell}$ -values are readily obtained by

$$(5) \quad W_{\alpha, \ell} = \sqrt{\ell/3} \Phi^{-1}(1 - \frac{1}{2}\alpha),$$

where $\Phi^{-1}(\alpha)$ is the inverse of the distribution function of the standard normal distribution. Miller used an approximation to derive (5) by noting that under H_0 , SR_n/n behaves like a Wiener process with zero drift and variance $(1/3)n$. A selection of $W_{\alpha, \ell}$ -values are given in Table I:

TABLE I

 $W_{\alpha, \ell}$ -values

α	$\ell=10$	15	20	25	30	40	50	100
0,005	5,13	6,28	7,26	8,11	8,89	10,26	11,47	16,22
0,01	4,71	5,77	6,66	7,45	8,16	9,42	10,53	14,90
0,025	4,09	5,01	5,78	6,47	7,08	8,18	9,15	12,93
0,05	3,58	4,38	5,06	5,66	6,20	7,16	8,00	11,32
0,10	3,00	3,68	4,25	4,75	5,20	6,01	6,72	9,50

To form an idea of the accuracy of the approximation, Monte Carlo trials were carried out, 500 times each, for some selected cases to estimate the probabilities of type I error:

$$\alpha_{\ell}^* = P_{\theta=0}(\max_{1 \leq n \leq \ell} |SR_n/n| \geq W_{\alpha, \ell})$$

and

$$\alpha_{\ell} = P_{\theta=0}(\max_{1 \leq n \leq \ell} (SR_n/n) \geq W_{\alpha, \ell}).$$

($\hat{\alpha}_{\ell}^*$ and $\hat{\alpha}_{\ell}$ are the estimated two-sided and one-sided levels of significance of the proposed test).

TABLE II

 $\hat{\alpha}_{\ell}^*$ -values for two-sided test

2α	$\ell=10$	15	20	25	30	35	40	45	50
0,05	0,062 (0,011)	0,066 (0,011)	0,054 (0,010)	0,046 (0,009)	0,046 (0,009)	0,040 (0,009)	0,048 (0,010)	0,048 (0,010)	0,040 (0,009)
0,01	0,008 (0,0040)		0,006 (0,0035)		0,008 (0,0040)		0,010 (0,0044)		0,014 (0,0053)

TABLE III
 $\hat{\alpha}_Q$ -values for one-sided test

α	$\ell=10$	15	20	25	30	35	40	45	50
0,05	0,046 (0,006)	0,043 (0,006)	0,050 (0,007)	0,048 (0,007)	0,045 (0,007)	0,039 (0,006)	0,046 (0,007)	0,039 (0,006)	0,039 (0,006)
0,025	0,006 (0,0035)		0,026 (0,0071)		0,016 (0,0056)		0,010 (0,0044)		0,010 (0,0044)

The values in brackets are the estimated standard deviations of the $\hat{\alpha}_Q^*$ and $\hat{\alpha}_Q$ -values. It is apparent that the approximation in the range $10 \leq \ell \leq 50$ for the two-sided test is quite good if the Monte Carlo-probabilities are considered as the real probabilities. The same can be said about the one-sided test for $\alpha = 0,05$ but it seems to be conservative for $\alpha = 0,025$. In the following, attention will be restricted to the two-sided case.

3. The performance of test in selected cases

Miller studied the properties of the test by looking at the OC-function and ASN-values under several values of θ for the double exponential distribution with density

$$f(x) = \frac{1}{2} \exp(-|x - \theta|), \quad -\infty < x < \infty.$$

He considered the cases where $\ell = 20$ and 50 both for one-sided and two-sided tests.

Since Miller gave no reason why a sequential test should be applied in a situation as described in Section 1, an attempt will now be made to show how this test performs in comparison with a corresponding fixed sample size procedure (the Wilcoxon-one-sample rank test). To do this, the OC-functions of the two procedures will be compared when the sample sizes are equal. Since the sequential test has a sample size which is stochastic, it seems reasonable to compare it with a fixed sample size procedure with sample size the same as the median sample size of the former procedure. This will be done for several θ - and ℓ -values. In addition to Miller, the case is also considered where the underlying distribution is uniform on the interval $[-1+\theta; 1+\theta]$. To form a better picture of the distribution of the stopping variable N in different cases, the first and third quartiles $Q_\theta^{(1)}(N)$ and $Q_\theta^{(3)}(N)$ are estimated, apart from the median $Me_\theta(N)$, the ASN $E_\theta(N)$ and standard deviation $\sigma_\theta(N)$.

The Tables IV and V give the results of Monte Carl trials which were performed 500 times on each case (only the two-sided case was studied):

TABLE IV

Double exponential distribution

$l = 20$

$ \theta $	OC		$E_{\theta}(N)$	$\sigma_{\theta}(N)$	$Q_{\theta}^{(1)}(N)$	$Me_{\theta}(N)$	$Q_{\theta}^{(3)}(N)$
	Sequential	Fixed Sample					
0,0	0,960(0,009)	0,950(0,010)	15,7	1,59	15	15	17
0,25	0,862(0,015)	0,896(0,014)	16,2	1,89	15	16	17
0,50	0,604(0,022)	0,644(0,021)	16,5	2,40	15	17	18
0,75	0,322(0,021)	0,384(0,022)	16,1	2,71	14	16	18
1,00	0,136(0,015)	0,246(0,019)	14,9	2,90	12	15	17
1,25	0,054(0,010)	0,144(0,016)	13,8	2,74	11	13	16

$l = 50$

0,0	0,960(0,009)	0,954(0,009)	43,7	2,53	42	43	45
0,25	0,730(0,020)	0,726(0,020)	43,4	5,18	42	44	47
0,50	0,222(0,019)	0,318(0,021)	37,8	8,50	31	38	45
0,75	0,008(0,004)	0,118(0,014)	29,6	7,25	24	29	35

TABLE V
Uniform distribution

$\ell = 20$

$ \theta $	OC		$E_{\theta}(N)$	$\sigma_{\theta}(N)$	$Q_{\theta}^{(1)}(N)$	$Me_{\theta}(N)$	$Q_{\theta}^{(3)}(N)$
	Sequential	Fixed Sample					
0,0	0,960*(0,009)	0,950 ⁺	15,7*	1,59*	15*	15*	17*
0,1	0,926 (0,012)	0,926(0,012)	15,9	1,65	15	16	17
0,2	0,790 (0,018)	0,816(0,017)	16,5	1,98	15	16	18
0,3	0,540 (0,022)	0,550(0,022)	16,7	2,21	15	17	18
0,4	0,322 (0,021)	0,378(0,022)	16,2	2,57	14	16	18
0,5	0,090 (0,013)	0,236(0,019)	15,1	2,64	13	15	17
0,6	0,024 (0,007)	0,100(0,013)	13,8	2,19	12	13	15
0,8	0,0	—	11,8	1,09	11	11	12

$\ell = 50$

0,0	0,960*(0,009)	0,950 ⁺	43,7*	2,53*	42*	43*	45*
0,1	0,842 (0,016)	0,820(0,017)	43,9	3,70	42	44	46
0,2	0,478 (0,022)	0,428(0,022)	41,5	7,08	38	43	47
0,3	0,106 (0,014)	0,210(0,018)	35,3	8,25	29	35	42
0,4	0,006 (0,003)	0,094(0,013)	28,8	6,29	24	28	32
0,6	0,0	—	21,8	2,93	20	21	23
0,8	0,0	—	18,8	1,03	18	18	19

Again the values in brackets are estimated standard deviations of the estimated OC-values. The table-values which are denoted by "*", are the corresponding values in Table IV, displayed here, while the value 0,95 (denoted by "+") is the theoretical value.

It follows from Table IV for $\ell = 20$, $|\theta| \geq 0,25$ that the OC-function of the sequential test has lower values than these of the comparable fixed sample test. This suggests that for far alternatives the sequential test gives a stronger power, or in other words: for the same power the typical sequential test would save observations under far alternatives. The same conclusions can be made for far alternatives in the case $\ell = 50$ in Table IV, and also for Table V in both cases. Monte Carlo results were also carried out in the case of a normal distribution and similar results were obtained.

To give an indication of the saving of observations of the sequential procedure in comparison with the fixed sample procedure, the following approximate method can be

used:

According to Lehmann (1975), pp. 165-166, the distribution of the Wilcoxon one-sample signed rank statistic can be approximated by a normal distribution for any θ and for n large. The OC-function of the test based on this statistic, would be

$$(6) \quad \gamma(\theta) = P_{\theta}(|SR_n| \leq c(\alpha, n)) \doteq \Phi(b_n(\theta)) - \Phi(a_n(\theta)) \text{ where}$$

$$(7) \quad b_n(\theta) = (c(\alpha, n) - E_{\theta}(SR_n) + \frac{1}{2}) / (\text{Var}_{\theta}(SR_n))^{\frac{1}{2}},$$

$$(8) \quad a_n(\theta) = (-c(\alpha, n) - E_{\theta}(SR_n) - \frac{1}{2}) / (\text{Var}_{\theta}(SR_n))^{\frac{1}{2}},$$

and $c(\alpha, n)$ is the critical value of the test, depending on α .

In the case where the underlying distribution is double exponential, (7) and (8) are obtained by noting that

$$E_{\theta}(SR_n) = n(n-1) \{ \frac{1}{2}(1 - (1+\theta)e^{-2\theta}) \} + n(1-e^{-\theta})$$

and

$$\begin{aligned} \text{Var}_{\theta}(SR_n) = & n(n-1)(n-2) \{ (1/3)(5e^{-2\theta} - (4 + 6\theta + 3\theta^2)e^{-4\theta}) \} \\ & + n(n-1) \{ (4+3\theta)e^{-2\theta} - 2(1+\theta)e^{-3\theta} - \frac{1}{2}(1+\theta)^2e^{-4\theta} \} \\ & + n(2e^{-\theta} - e^{-2\theta}), \end{aligned}$$

while for the uniform case

$$E_{\theta}(SR_n) = n(n-1) \cdot \frac{1}{2}\theta(2-|\theta|) + n\theta,$$

$$\begin{aligned} \text{Var}_{\theta}(SR_n) = & n(n-1)(n-2) \{ (1/3)(1-6\theta^2+8|\theta|^3-3\theta^4) \} \\ & + n(n-1) \{ \frac{1}{2}(3-10\theta^2+8|\theta|^3-\theta^4) \} + n(1-\theta^2). \end{aligned}$$

For fixed $\theta > 0$, assuming that the argument $a_n(\theta)$ in (6) is negligible, n can be solved iteratively by the Newton-Raphson method for given $\gamma(\theta)$ -value:

$$(9) \quad n_{r+1} = n_r - f(n_r)/f'(n_r), \quad r = 1, 2, \dots$$

where

$$f(n) = c(\alpha, n) - E_{\theta}(SR_n) + \frac{1}{2} - \Phi^{-1}(\gamma(\theta)). (\text{Var}_{\theta}(SR_n))^{\frac{1}{2}}.$$

$f'(n)$ is the derivative of $f(n)$ with respect to n .

(The case $\theta < 0$ is symmetric to the case $\theta > 0$; therefore only the latter case will be considered.)

The initial value n_1 is chosen as the median sample number of the corresponding sequential procedure. Tables VI and VII show the sample sizes (accurate to one decimal place) which are necessary to give the same OC-values as those of the corresponding sequential test. For $\ell = 20$ values of $a(\theta) = \max_{10 \leq n \leq 50} a_n(\theta)$ are tabulated while for $\ell=50$ values of $a(\theta) = \max_{20 \leq n \leq 50} a_n(\theta)$ are given. These $a(\theta)$ -values provide the necessary motivation for neglecting the term $\Phi(a_n(\theta))$ in (6) when solving for n . In the second last column of the tables, the P-values give the approximate OC-values (calculated from (6)) for the fixed-sample test. (For each value of θ the sample size is taken equal to the median sample size of the sequential test given in Tables IV and V.) The last columns of the tables give the $E_{\theta}(N)$ -values obtained in Tables IV and V.

Comparison of the P-values with the Monte Carlo OC-values given in the third column of Tables IV and V, seems to indicate that the normal approximation in all cases considered is quite good. Another conclusion which can be made when comparing the $E_{\theta}(N)$ -values with the n -values, is that for far alternatives the saving of observations of the sequential test is considerable (as much as 25% in the case $\ell = 20$ and 44% in the case $\ell = 50$).

TABLE VI
Double exponential case, $2\alpha = 0,05$

	$ \theta $	n	$a(\theta)$	P	$E_{\theta}(N)$
$\ell = 20$	0,25	19,1	-2,783	0,870	16,2
	0,50	22,2	-3,663	0,668	16,5
	0,75	21,2	-4,706	0,440	16,1
	1,00	19,0	-5,977	0,261	14,9
	1,25	17,1	-7,545	0,179	13,8
$\ell = 50$	0,25	46,4	-3,029	0,723	43,4
	0,50	47,1	-4,193	0,318	37,8
	0,75	52,8	-5,559	0,132	29,6

TABLE VII

Uniform case, $2\alpha = 0,05$

	$ \theta $	n	$a(\theta)$	P	$E_{\theta}(N)$
$q = 20$	0,1	16,2	-2,572	0,914	15,9
	0,2	19,3	-3,223	0,806	16,5
	0,3	20,0	-4,018	0,607	16,7
	0,4	19,2	-5,056	0,412	16,2
	0,5	20,1	-6,491	0,235	15,1
	0,6	17,3	-8,628	0,132	13,8
$q = 50$	0,1	44,2	-2,776	0,819	43,9
	0,2	43,0	-3,667	0,477	41,5
	0,3	48,3	-4,762	0,232	35,3
	0,4	47,2	-6,195	0,098	28,8

4. The OC-function and ASN-values obtained by a Wiener process approximation

Miller gave an approximation of the OC-function, based on the assumption that under $\theta \neq 0$, $Y_n = SR_n/n$ is also behaving like a Wiener process; this time with drift $\mu(\theta)n$ and variance $A^2(\theta)n$, where $\mu(\theta)$ and $A(\theta)$ are given by

$$E_{\theta}(Y_n) = \mu(\theta)n + o(1)$$

and

$$\text{Var}_{\theta}(Y_n) = A^2(\theta)n + o(1) \text{ when } n \rightarrow \infty$$

where $\mu(\theta)$ and $A^2(\theta)$ are the coefficients of n^2 and n^3 in the expressions for $E_{\theta}(SR_n)$ and $\text{Var}_{\theta}(SR_n)$ respectively (see (7) and (8)).

To give an approximation to the OC-function in the two-sided case, it is necessary to evaluate the expression

$$(10) \quad \gamma_q(\theta, \alpha) = 1 - P_{\theta} \left(\max_{0 \leq t \leq q} |W(t) + \mu(\theta)t/A(\theta)| \geq W_{\alpha, q/A(\theta)} \right), \text{ where } W(t) \text{ is a standard Wiener process.}$$

Samuel-Cahn (1974) gave an expression (equation (2.5)) obtained from Anderson (1960) to evaluate $\gamma_q(\theta, \alpha)$:

$$(11) \quad \gamma_{\ell}(\theta, \alpha) = 1 - [\phi(\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta)) + \phi(\delta(\theta)\ell^{1/2} - W_{\alpha}\sigma(\theta))].$$

$$\sum_{j=0}^{\infty} (-1)^j \exp \{-2j(j+1)W_{\alpha}^2\sigma^2(\theta)\} [R((2j+1)W_{\alpha}\sigma(\theta) + \delta(\theta)\ell^{1/2}) + R((2j+1)W_{\alpha}\sigma(\theta) - \delta(\theta)\ell^{1/2})],$$

$$\text{where } \delta(\theta) = \mu(\theta)/A(\theta), W_{\alpha} = \Phi^{-1}(1-\alpha/2),$$

$$\sigma(\theta) = 1/(3^{1/2}A(\theta)) (= A(0)/A(\theta)),$$

and $\phi(x)$ is the density function of a standard normal distribution, while

$$R(x) = (1 - \Phi(x))/\phi(x)$$

is Mill's ratio.

In practice only the first two terms in the infinite series in (11) will be used, since the contribution of the remainder is negligible. The approximation to the OC-function is therefore:

$$(12) \quad \gamma_{\ell}^*(\theta, \alpha) = 1 - \{R(W_{\alpha}\sigma(\theta) + \delta(\theta)\ell^{1/2}) + R(W_{\alpha}\sigma(\theta) - \delta(\theta)\ell^{1/2})\} \\ + \exp \{-4W_{\alpha}^2\sigma^2(\theta)\} [R(3W_{\alpha}\sigma(\theta) + \delta(\theta)\ell^{1/2}) + R(3W_{\alpha}\sigma(\theta) - \delta(\theta)\ell^{1/2})] \} \\ [\phi(\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta)) + \phi(\delta(\theta)\ell^{1/2} - W_{\alpha}\sigma(\theta))].$$

Miller considered the double exponential case but he used a more crude approximation to (10). For this case:

$$\mu(\theta) = \frac{1}{2}[1 - (1 + \theta)e^{-2\theta}], A^2(\theta) = (1/3)[5e^{-2\theta} - (4 + 6\theta + 3\theta^2) \cdot e^{-4\theta}],$$

while in the uniform case:

$$\mu(\theta) = \frac{1}{2}\theta(2 - |\theta|) \text{ and } A^2(\theta) = (1/3)(1 - 6\theta^2 + 8|\theta|^3 - 3\theta^4).$$

Using equations (2.6) and (2.7) of Samuel-Cahn (which are obtained from Anderson), approximations (again by using the first two terms in an infinite series) to the ASN-values are:

$$\begin{aligned}
 (13) \quad \eta_{\ell}^*(\theta, \alpha) = & \ell - [\phi(W_{\alpha}\sigma(\theta) - \delta(\theta)\ell^{1/2}) + \phi(W_{\alpha}\sigma(\theta) + \delta(\theta)\ell^{1/2})] \\
 & \cdot \ell/(\delta(\theta)\ell^{1/2})[(\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta))R(\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta)) \\
 & - (-\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta))R(-\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta)) \\
 & - \exp(-4W_{\alpha}^2\sigma^2(\theta)) \{(\delta(\theta)\ell^{1/2} + 3W_{\alpha}\sigma(\theta))R(\delta(\theta)\ell^{1/2} + 3W_{\alpha}\sigma(\theta)) \\
 & - (-\delta(\theta)\ell^{1/2} + 3W_{\alpha}\sigma(\theta))R(-\delta(\theta)\ell^{1/2} + 3W_{\alpha}\sigma(\theta))\}] \text{ for } \theta \neq 0
 \end{aligned}$$

and

$$\begin{aligned}
 (14) \quad \eta_{\ell}^*(0, \alpha) = & \ell - 4\ell\phi(W_{\alpha})[(1+W_{\alpha}^2)R(W_{\alpha}) - W_{\alpha} \\
 & - \exp(-4W_{\alpha}^2) \{(1+9W_{\alpha}^2)R(3W_{\alpha}) - 3W_{\alpha}\}].
 \end{aligned}$$

The values of $\eta_{\ell}^*(\theta, \alpha)$ give approximations of an upper bound of $E_{\theta}(N)$, since it does not take into consideration the option to stop when (4) is violated.

Tables VIII and IX give calculated values for $\gamma_{\ell}^*(\theta, \alpha)$ and $\eta_{\ell}^*(\theta, \alpha)$ in selected cases:

TABLE VIII

Double exponential distribution, two-sided test at level $2\alpha=0,05$

θ	$\ell=20$		$\ell=50$	
	$\gamma_{\ell}^*(\theta, \alpha)$	$\eta_{\ell}^*(\theta, \alpha)$	$\gamma_{\ell}^*(\theta, \alpha)$	$\eta_{\ell}^*(\theta, \alpha)$
0,0	0,950(0,945)	19,8	0,950(0,939)	49,4
0,25	0,863	19,4	0,711(0,68)	46,5
0,50	0,642(0,63)	18,3	0,226(0,20)	38,1
0,75	0,373	16,9	0,020(0,02)	29,9
1,00	0,154(0,14)	15,4	0,0 (0,0)	25,1
1,25	0,037	14,1	0,0	22,4

TABLE IX

Uniform distribution, two-sided test at level $2\alpha=0,05$

	$\ell=20$		$\ell=50$	
$ \theta $	$\gamma_{\ell}^*(\theta, \alpha)$	$\eta_{\ell}^*(\theta, \alpha)$	$\gamma_{\ell}^*(\theta, \alpha)$	$\eta_{\ell}^*(\theta)$
0,0	0,950	19,8	0,950	49,4
0,1	0,900	19,6	0,807	47,8
0,2	0,764	19,0	0,437	42,5
0,3	0,558	18,0	0,105	35,0
0,4	0,311	16,7	0,005	28,6
0,5	0,096	15,2	0,0	24,4
0,6	0,006	13,8	0,0	21,8

The values in brackets are the corresponding (simpler) Wiener approximations which Miller used, which do not differ much from those calculated above. Comparing Tables VIII and IX with IV and V, it seems reasonable to believe that $\gamma_{\ell}^*(\theta, \alpha)$ gives a good indication of the OC-function for the two-sided test. For far alternatives $\eta_{\ell}^*(\theta, \alpha)$ also approximate the ASN-values quite accurately — in practice these values are more important. The approximation which Miller used can be implemented to approximate the OC-function in the one-sided test at level α :

$$(15) \quad \tilde{\gamma}_{\ell}(\theta, \alpha) = 1 - \exp(2W_{\alpha}\delta(\theta)\ell^{1/2}\sigma(\theta))\Phi(-\delta(\theta)\ell^{1/2} - W_{\alpha}\sigma(\theta)) - \Phi(\delta(\theta)\ell^{1/2} - W_{\alpha}\sigma(\theta)).$$

From Samuel-Cahn the following approximations are obtained for the ASN-values in the one-sided test at level α (equation (1.15))

$$(16) \quad \begin{aligned} \tilde{\eta}_{\ell}(\theta, \alpha) = & \ell[\Phi(-\delta(\theta)\ell^{1/2} + W_{\alpha}\sigma(\theta)) \{1 - W_{\alpha}\sigma(\theta)/(\delta(\theta)\ell^{1/2})\} \\ & - \Phi(-\delta(\theta)\ell^{1/2} - W_{\alpha}\sigma(\theta))\exp(2\delta(\theta)\ell^{1/2}\sigma(\theta)) \{1 + W_{\alpha}\sigma(\theta)/(\delta(\theta)\ell^{1/2})\} \\ & + W_{\alpha}\sigma(\theta)/(\delta(\theta)\ell^{1/2})] \text{ for } \theta \neq 0, \end{aligned}$$

and

$$(17) \quad \tilde{\eta}_{\ell}(0, \alpha) = \ell[1 - 2\alpha(W_{\alpha}^2 + 1) + 2W_{\alpha}\phi(W_{\alpha})].$$

6. Practical application of Miller's test

The purpose for studying the properties of the test in selected cases, was to recommend its use with greater confidence. In the range $k=20$ to 50 the OC-function as well as the ASN-values in the cases studied suggest the general use of the test. The simplicity of the stopping rule, and the sole assumption of symmetry concerning the underlying distribution of the population, make the test practically appealing.

In practice the significance-level and truncation point can be decided upon after taking in consideration ethical, safety and economical principles. To form an idea of the probability of type II error and the ASN at a given two sided alternative θ -value, (12) and (13) can be used for an assumed underlying distribution. For the one-sided case (15) and (16) can be used analogously. In cases where the alternative differs little from zero and where $\sigma(\theta)$ differs little from unity, Table V of Samuel-Cahn can also be used with her $\mu = \delta(\theta)k^{1/2}$ (For $\mu > 2,8$, her Table I is applicable.)

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